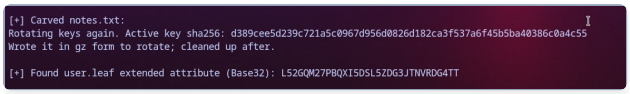
Fallen Leaf

Digital Forensics



I extracted `leafdisk.img` from `leafdisk.img.xz`. Rather than manually carving files, I used `binwalk -Me` to recursively extract all embedded compressed streams.

In the extracted filesystem root, I found `flag.txt.enc` alongside bash history referencing an OpenSSL encryption command.

Inspecting `.bash_history` revealed the exact encryption command: `openssl enc -aes-256-cbc -d -in flag.txt.enc -pass pass:AutumnLeaves2026!`. Executing this command with OpenSSL decrypted the flag immediately.

Finally, I tested the recovered flag against the executable / service:

`python3 Fallen_Leaf.py`

OpenSSL AES-256-CBC decryption verified.

Flag Captured

`SecLeaf{3xtr4ct_4ll_th3_f4ll3n_l34v35}`

BY:JOKER_53